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Evolutionary analysis of the collaboration networks within National Quality Award Projects of China

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Abstract

Construction organizations are increasingly cooperating in what are often referred to as "collaboration network" that enabled them to share risks, pool resources and explore opportunities to jointly participate in managing large-scale projects. In this study, we construct the so-called inter-contractors' collaboration networks by mapping the electronic database of NQAPC for an 8-year period (2003–2010). In these networks, nodes represent contractors; two contractors are connected by an edge if they have cooperated at least one project. By using a variety of network measures, i.e., giant component, degree distribution, average path length, and clustering coefficient, we aim to descriptively investigate the structural evolution of the collaborations between contractors in the construction industry of China. As network size increases, we find a structural transition in the collaboration community size, the degree follows power-law distribution with an exponential cutoff, the average path length tends to decrease, and the clustering coefficient slightly decreases. Some explanations and a series of construction insights are discussed. The results and methodologies not only would help governors understand the social mechanisms underlie processes of construction industry, but could help contractors choose competent partners by identifying network properties.
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Keywords: Collaboration network; National Quality Award Project of China (NQAPC); Contractors; Project management; Construction industry

1. Introduction

The last few years have witnessed a tremendous activities devoted to the understanding of complex networks, such as the World Wide Web (Albert et al., 1999), metabolic networks (Stelling et al., 2002), food webs (Williams et al., 2002), and last but not least, social networks (Liljeros et al., 2001; Wasserman and Faust, 1994). A social network is considered a social structure composed of actors that are connected by one or more specific types of relationships, e.g., friendships, firm alliances, and international trades (Liljeros et al., 2001; Newman, 2001a; Wasserman and Faust, 1994). As an important research subject, social network analysis (SNA) has so far centered on intra- and inter-organizational relationships, i.e. within and between organizations. The former assess the level and performance of partnerships between individuals within an organization, while the latter assess the level and performance of partnerships between organizations.

In construction field, all organizations are social networks and can be addressed in terms of a set of nodes linked by social relationships, i.e., intra-organizational relationships (e.g., Pryke, 2004). Further, the environment in which an organization operates might be viewed as a network of other organizations, i.e., inter-organizational relationships (Pryke, 2004). Focusing on intra-organizational relationships, Chinowsky et al. (2008, 2010) introduced social network model into project organization development, and developed high-performance organizations through an analysis of both leadership and team actions within a

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project team framework. Di Vincenzo and Mascia (2012) defined social capital in project-based organization as the way individuals are linked, and provided evidence that levels of project performance are significantly associated with the particular structure of projects' social capital. Using centrality measures of SNA, Hossain and Wu (2009), and Hossain (2009) explored the association between communications network centrality and coordination score from a real-world dataset of complex projects with hundreds of people working together. Chowdhury et al. (2011) applied network theory to identify and distinguish potential stakeholders in Public Private Partnership (PPP) affiliation.

In the inter-organizational networks, actors (nodes) are identified as construction firms, rather than individuals. In this respect, Pryke (2004, 2005) proposed that the project coalition be conceptualized as a network of relationships. The methodology was applied to investigate the managerial attributes of UK construction projects with regard to procurement modes. Using a social network analysis approach, Park et al. (2011) produced a collaboration network model to investigate a variety of collaboration patterns and their impacts on the level of profit performance. Recently, Son and Rojas (2011) introduce an agent-based simulation of the evolution of collaboration within inter-organizational networks of construction project teams. In each of the cases, quantitative indicators of SNA offer potentials for guiding researchers and practitioners in monitoring the organization characteristics (e.g., Hossain, 2009; Hossain and Wu, 2009; Pryke, 2004). However, there are limitations in previously mentioned social network studies in construction field. Firstly, most of the networks constructed in SNA studies were identified through interviewing participants and distributing questionnaires (e.g., Pryke, 2005). Secondly, most of the SNA analyses focused on the ego-network properties, e.g., core-peripheral stakeholders, influential intermediary participants and their interdependence (e.g., Chinowsky et al., 2010; Chowdhury et al., 2011; Di Vincenzo and Mascia, 2012). Although these studies have revealed details about the cognitive, psychological, and sociological features of the networks, they suffer from several problems (e.g., subjectiveness, sampling issues, and statistical accuracy) and limit themselves to small-size networks—typically to tens of stakeholders or people (e.g., Chinowsky et al., 2008; Pryke, 2004). Last and foremost, most of the studies focused on the frozen network in a given year (e.g., Chowdhury et al., 2011). However, the social network constantly expands by the addition of new actors and acts.

A major class of social networks is the collaboration network, defined as a network of actors (e.g., movie stars or paper authors) involved in collaboration acts (e.g., movies or scientific papers). Collaboration networks are considered as an important research subject in both social network studies (Wasserman and Faust, 1994) and complex network investigations (Barabási and Albert, 1999; Watts and Strogatz, 1998), such investigations including movie actor collaboration networks (Zhang et al., 2006; Chang et al., 2007), scientific collaboration networks (Newman, 2001a, 2001b, 2001c), and so on. Newman (2001a,b, 2004) has taken an important step towards applying complex network measures to reveal the global structural properties of scientific collaboration networks for biomedicine, physics, and mathematics. In these networks, nodes are scientists, two scientists are connected by an edge if they have wrote a paper. The collaboration network research provides a powerful paradigm for the representation and analysis of collaboration relationships between organizations. More specifically, the collaboration network research mainly focused on the global structure of the network, and the network constructed is large—containing over a million individuals—and for which a reliable statistical description of the topological properties is possible (Newman, 2001a, 2001b).

As construction projects are getting larger and more complex, no single contractor with a limited capacity and partial information can handle all matters (Son and Rojas, 2011). The importance of collaboration, coordination and communication for effective project execution has been emphasized in the construction industry (de Saram and Ahmed, 2001; Morton et al., 2006; Shohet and Frydman, 2003). A number of collaborative forms including partnerships, strategic alliances and joint ventures (Alvarez and Barney, 2001; Anvuur and Kumaraswamy, 2007; Badger and Mulligan, 1995; Ozorhon et al., 2008; Walker and Johannes, 2003; Wong and Cheung, 2005; Xu et al., 2005) suggest that collaboration among heterogeneous construction contractors is increasing in a wide range of construction fields. One of the most important motivations for forming collaboration among contractors is that collaboration can provide sufficient financing strength to participate in capital-hungry projects. Other motivations include bridging knowledge and expertise gaps, sharing of the burdens of risk exposure, and exploring local market opportunities (Walker and Johannes, 2003).

A comprehensive investigation of how contractors gathered and evolved from a global network view can not only illuminate the development trend of organizations' collaborations in the industry level, but also provide evidence of better performance for collaborative entities and thus prove beneficial when firms develop their strategies to improve project performance at the firm level (Pryke, 2004). To our knowledge, few studies have attempted to assess inter-organization collaboration relationships in construction field from an industry perspective. Focusing the contractors' collaborations in the National Quality Award Projects of China (NQAPC), we aim to construct and present analysis of the global structural properties of realistic inter-organizational collaboration networks (containing hundreds of heterogeneous contractor companies, corporations and groups). The relatively larger collaboration networks within NQAPC permit to obtain solid statistic conclusions concerning the global structure. More specifically, our purpose is to empirically investigate the structure of collaboration patterns in the construction field in China and explore the evolutionary characteristics by analyzing real cases of collaboration at first time. Our goal is to extract main properties and mechanisms that are crucial to the understanding of collaboration progress within NQAPC from a global network view. This study goes one step further than previously mentioned SNA studies by providing a comprehensive and systematical picture of the principle of how inter-organizational network evolved from an industry perspective.

The outline of the paper is as follows. In Section 2 we present the main idea for constructing the NQAPC contractors'

collaboration network. In Section 3 we focus our attention on topological properties such as degree distribution, clustering coefficient, and shortest path distribution, on the analysis of the time evolution, especially the behavior of the largest connected component in the network. In Section 4 we discuss the findings and implications for the global evolution of the China construction field. The study is concluded with the most important meaningful findings and implications that may quote some generic value to readers from other developing countries.

2. The contractors' collaboration networks within NQAPC

Generally speaking, collaboration networks can be described using bipartite graphs with two types of nodes (Albert and Barabási, 2002; Ramasco et al., 2004). One type of nodes can be called "actor" such as firms or individuals. Another type of nodes can be called "act" such as projects or activities. The edges between different kinds of nodes denote an affiliation relationship, i.e., an actor is taking part in an act. If one considers one type of nodes only, two edges sharing a common node in the bipartite graph are projected onto an edge between the two nodes of the same type. If two actors cooperate in more than one act, the relation can be expressed by multiple edges between them. As a rule, it is the projections (unipartite) of these bipartite graphs that are empirically studied. In these projections, the nodes representing the "acts" are excluded, and collaborating pairs of "actors" are connected by edges (Newman, 2001b).

The Nation Quality Award (since 1983) is one of the highest quality awards in the China construction field. The award-winning projects serve as the representative samples of the national critical construction project population. The award-winning participants are owners, supervisors, planners and contractors devoted to the construction of national critical infrastructures (http://www.cacem.com.cn/). This study mainly investigates contractors' collaboration network within NQAPC by mapping the electronic database containing about eight hundred contractors for an 8-year period (2003–2010). A projected one-mode (unipartite) network of the bipartite network is used.

Fig. 1 shows a schematic view for explaining bipartite graph and its projection where the squares labeled by *P1*, *P2* and *P3* are "acts" (projects); the circles labeled by *C1–C9* are "actors" (contractors). The projects can be performed by one contractor (solo), two contractors (double), and three or more contractors (group). In the projection all contractors participating in one project are connected by edges. This seems a reasonable definition of construction collaboration: most contractors who have cooperated within a project will know one another quite well. Thus, each project is represented by a "project complete subgraph (*PCS*)". Some contractors are shared among different projects (the contractors shared by two projects are shown by darker black circles) and can be seen as bridges between different *PCSs*. After sorting 1-year NQAPC this way, an undirected unipartite graph, i.e., the set of *PCSs*, can finally be formed for contractors' collaboration network each year.

Before proceeding we clarify a few methodological issues that affect the data analysis. First, in the database, contractors belonging to the same construction group will appear to be different nodes in the network, as long as they are contractors with "Super Grade" or "Grade A" qualifications, or specialized contractors with "Grade A" qualifications. These qualifications are regulated by the Administration of Construction Enterprise Qualifications proposed by the Ministry of Housing and Urban–Rural Development of the People's Republic of China (MOHURD) (http://www.mohurd.gov.cn/). High level qualification means that the contractors are capable of constructing larger-scale and more complex projects. For example, China Railway No.12, No.14, No.18 and No.19 Bureau Group Corporation Limited are all subordinated to China Railway Construction Corporation Limited, one of the largest construction enterprises in China (http://english.crcc.cn/). However, these four construction enterprises are all contractors with "Super Grade" qualifications, thus considered as different nodes in the collaboration networks.

Second, collaboration relationships between every pair of contractors are considered equivalent in a project. In order to characterize the strength of the interaction, one could attribute a

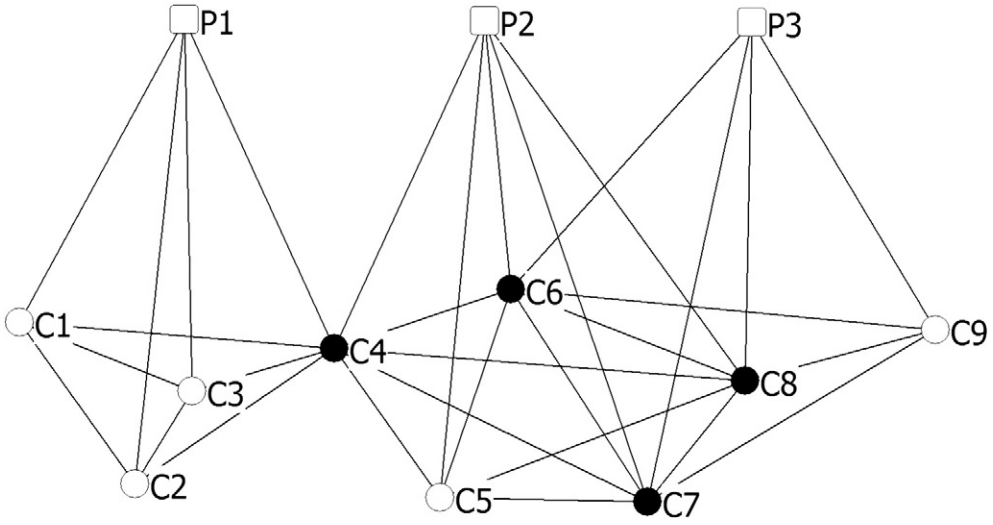


Fig. 1. Schematic view for explaining bipartite graph and its projection.

weight w_{ij} to an edge e_{ij} with a value proportional, say, to the number of projects cooperated by contractor i and j. However, in this study we use the simple unweighted network.

Third, in order to get information on structural evolution of the contractors' collaboration network, one needs a complete dataset of the constructed projects, ideally from the birth of the National Quality Award until today. However, web database does not cover so much of the past several years. Thus our study is limited to only a recent segment of the database (8-year period from 2003 to 2010 inclusive).

In this paper, the NQAPC data was "sliced" to eight separate networks, each representing the cumulative collaborative efforts of contractors within NQAPC up to an awarding year (2003–2010). Table 1 summarizes fundamental statistics for the period 2003 to 2010 under study. The first two rows of the table contain the number of the award-winning projects and contractors each year, respectively. When analyzed over time, the number of projects and contractors each year grows steadily over time, and the average number of contractors per project each year is almost the same (=2). The following two rows show the number of the award-winning projects and contractors up to the year 2004–2010, respectively. At the end of year 2010, the NQAPC database consists of 917 projects and 767 contractors. The last two rows show respectively the number of nodes (contractors) and edges (collaborations) in the cumulative largest component of contractors' collaboration network up to the given year (2003–2010). A component is defined as a subset of nodes interconnected by edges. To the end of the year 2010, the contractors' network within NQAPC was composed of a total of $N = 599$ connected nodes, i.e. contractors who have at least one collaborator, and a total of $E = 3359$ edges (collaborations) in the largest component. Fig. 2 plots the schematic view of the cumulative largest component in collaborators' collaboration networks up to the given year (2003–2010), giving us a good understanding of the evolution of the range and configuration of the networks over time.

3. Evolution of the contractors' collaboration networks

In this section, we will study the structural evolution of the contractors' collaboration network within NQAPC. In particular, we will find how the largest component forms and develops, and characterize the structural evolution of the largest components by several time varying global network measures. Such global network quantities are the average degree, the degree distribution, the average path length, and the clustering coefficient (Barabási and Albert, 1999; Newman, 2001b; Watts and Strogatz, 1998). For the largest component in 2003 is rather small, global quantities for the collaboration network in 2003 are meaningless and thus excluded.

3.1. The largest component increases and a giant component emerges

It is important to realize that the collaboration network is fragmented into many independent components. For example, until 2010 there are 108 independent components and 22 isolated nodes (a kind of component that has only one node and no edge) in the network, which means there are a very small fraction of isolated contractors that do not collaborate. There are several reasons for this. First, the contractors within one component are the only contractors of the project on which they appear to the extent of cooperating with a project. Second, in every field there are contractors who almost never collaborate with other contractors from other fields. This may correspond to technology boundaries, or geographical boundaries, or both. Fig. 3 shows the evolution of the number of components together with the number of isolated nodes present in the cumulative networks up to a given year. Strikingly, it is found that both the number of components and isolated nodes increase linearly. Also, the increase in the number of components parallels the increase in the number of isolated nodes. This indicates that each year there are a fixed number of award-winning projects which are constructed by independent contractors or isolated contractors.

In all social collaboration networks, with low number of connections between individuals one can easily find islands of collaborations. However, with the increase of the total number of connections there comes a point at which a giant component emerges—the largest group of actors who are all connected to one another by paths of intermediate relationships (Newman, 2001a). To investigate the emergence of this giant component in the contractors' collaboration network, we measured the relative size of the largest component, s, giving the ratio between the number of nodes in the largest component and the total number of nodes in the whole network, as time goes by year.

Fig. 4 presents the evolution of the largest component expressed in the value s. It is shown that s highlights two distinct time periods: a first term (2003–2007) during which s increases sharply from about 0.1 (2003) to 0.7 (2007), and a second term (2007–2010), during which the growth slows

Table 1
Fundamental statistics of the contractors' collaboration network within NQAPC in the period 2003–2010.

Quantity/year	2003	2004	2005	2006	2007	2008	2009	2010
Number of projects each year	54	74	86	116	155	138	149	145
Number of contractors each year	91	142	158	201	237	210	296	273
Number of projects up to a given year	54	128	214	330	485	623	772	917
Number of contractors up to a given year	91	207	296	400	500	590	680	767
Number of nodes in the largest component up to the given year (as a percentage %)	7 (7.7%)	47 (22.7%)	77 (26%)	198 (49.5%)	334 (66.8%)	434 (73.6%)	509 (74.9%)	599 (78.1%)
Number of edges in the largest component up to the given year	10	285	417	1228	1903	2354	2847	3359

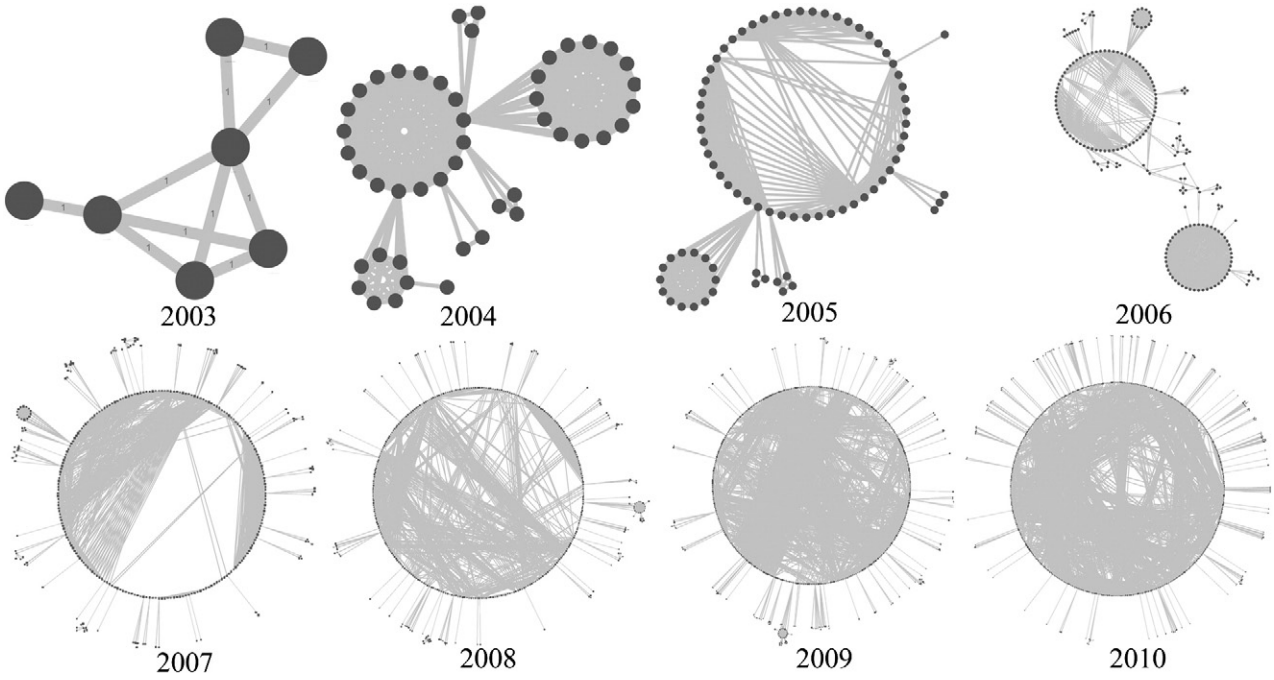


Fig. 2. Schematic view of the cumulative largest component in contractors' collaboration networks in the period 2004–2010.

down from 0.7 (2007) to 0.8 (2010). The fast convergence of the component size to an approximately stationary value around 0.7 indicates the emergence of the giant component, which means the basic collaboration alliances among contractors are uncovered by the end of 2007. Fig. 4 also shows the evolution of the relative size of the second-largest group of connected contractors. It is found that the relative size of the second-largest component increases for the first 3 years and then decreases steadily in the following years. In fact, as the collaboration networks increase in size, contractors may be gaining access to cooperate both directly (from the contractors with whom they cooperate) and indirectly (through the collaborators of their collaborators). In this way many contractors in smaller components are led to be part of the largest growing component.

In our case, the size of the giant component in 2010 is 599 contractors, which represents 78.1% of the total number of the construction contractors. The existence of the giant component has a social meaning. It means 78.1% of the contractors within NQAPC are members of a single community and every contractor in the community is connected to each other by some direct and indirect collaborations. In other words, it appears that the contractors' collaboration networks are very highly connected, and the construction market in China is in no immediate danger of fragmentation.

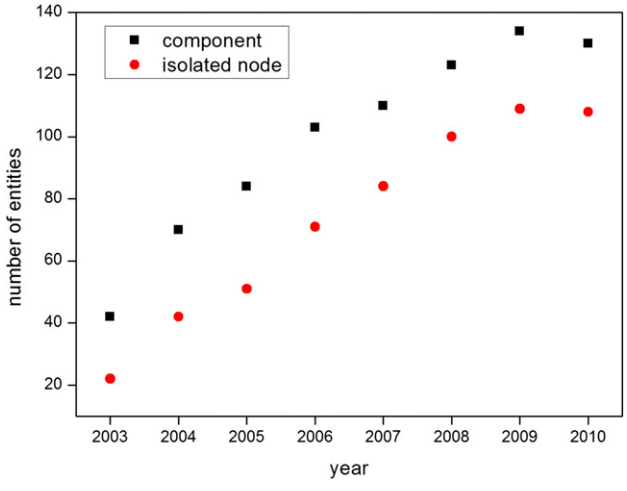


Fig. 3. Evolution of the cumulative number of components and isolated nodes in the network in the period 2003–2010.

3.2. Averge degree fluncuates

In collaboration network, a quantity characterizing the whole network's connectedness is the average degree <*k*>, giving the average number of edges (collaborations) per node (contractor). The average degree can reflect the compactness of the network. The higher the average degree is, the tighter the network (Newman, 2001a). The average degree of the network until 2010 is 11.21, meaning that each contractor, on average, has cooperated with more than eleven contractors. One can say that the collaboration between NQAPC contractors is tight.

Fig. 5 shows the time dependence of <*k*> for the cumulative largest components in the contractors' collaboration networks to a given year. It is found that <*k*> fluctuates during the first years but increase linearly after 2007 in which the giant component appears. It is also interesting to remark in Fig. 5 that <*k*> jumps from 11.5 to 11 at the emergence of the giant component. Then, the average degree increases with time indicating that, on average, more edges (collaborations) are added to the network than nodes (contractors). As confirmed by

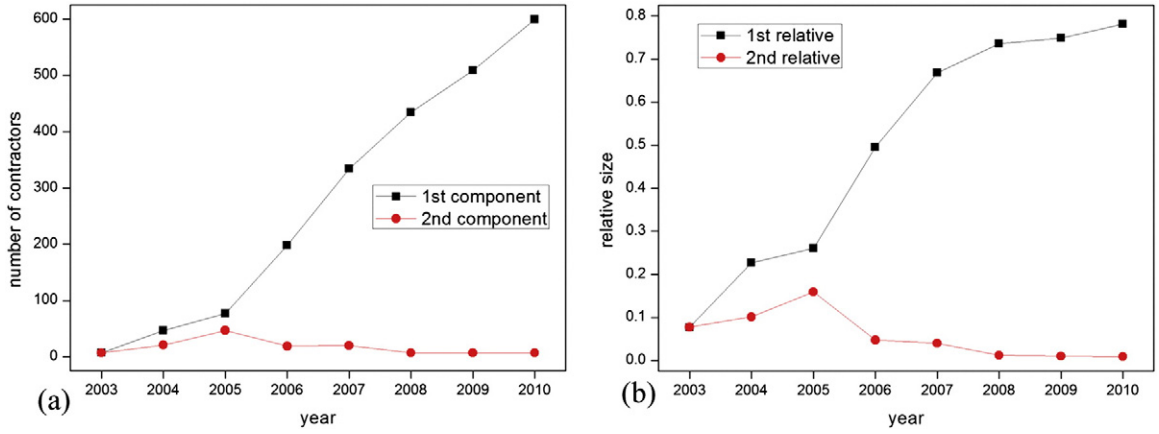


Fig. 4. Evolution of the cumulative relative size of the largest and second largest components in the period 2003–2010.

Fig. 6, which shows the number of nodes and edges of the largest component in the period 2004–2010. With time the number of nodes increases due to arrival of new contractors, the total number of edges grow faster than contractors as new edges come not only from collaborations involving new contractors, but also from contractors who were already in the collaboration network. And the collaborations made between old contractors are more often than the collaborations made by new contractors with old ones (seen in Fig. 9).

3.3. Degree distribution follows the same rule

A typically global network property that has been much investigated is the distribution of node degrees, *P(k)*, giving the probability that a randomly selected node (contractor) has k adjacent edges (direct collaborations). Networks for which *P(k)* has a power-law tail, $p(k) \sim k^{-\lambda}$, are known as scale-free networks (Barabási and Albert, 1999). In scale-free networks, most of the nodes (the trivial many) have low degrees while a small but significant fraction of nodes (the vital few) have an extraordinarily high degree. On the other hand, classical network models, including the Erdős-Rényi and the Watts-Strogatz (Watts and Strogatz, 1998) models have an exponentially decaying *P(k)* and are collectively known as exponential networks. Degree distribution describes the network structure and may imply the network evolution mechanism. For example, it is widely acknowledged that many real networks showing power-law degree distribution are induced by linear preferential evolution mechanism (Barabási and Albert, 1999).

Since the graph of the distribution $P(k)$ of the number k of collaborators per contractor is rather bumpy, especially in the tail of the distribution. Instead, we adopted the cumulative form, which is related to the original degree distribution through the following formula, $P(k>n) = \int_n^{k_{\max}} P(k)dk$. One advantage of using the cumulative form is that it can effectively reduce the number of statistical errors caused by the system size, while at the same time the shape of the distribution is well preserved (Newman, 2003).

Fig. 7 presents the $P(k > n)$ for the cumulative contractors' collaboration networks to a given year. Graphing on the double-log scale reveals the degree distributions do not appear

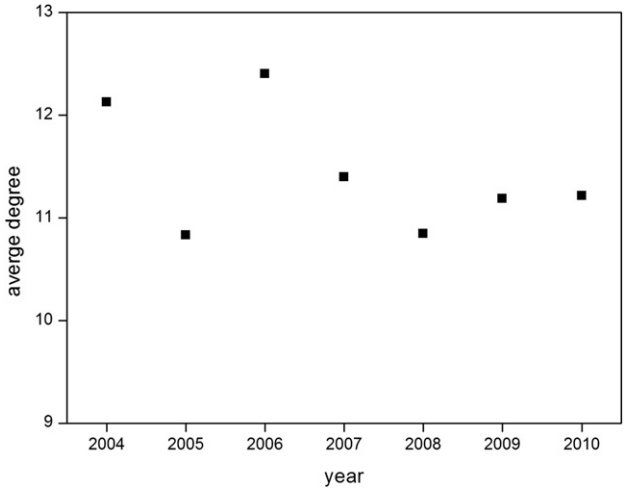


Fig. 5. Evolution of the average degree of the cumulative largest component up to the given year.

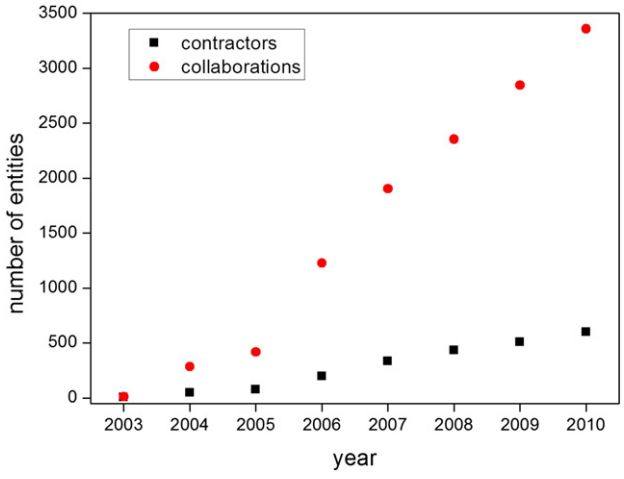


Fig. 6. Evolution of the number of contractors and collaborations of the largest component up to the given year.

to be a pure power-law. If they did, the curves in Fig. 7 would be straight lines on the logarithmic scales used. However, these data are well fitted by a power-law form with an exponential cutoff. For example, the observed degree distribution of the collaboration network until 2010 shows a power-law regime in the first part followed by an exponential decay in the tail:

$$p(k>n) \sim k^{-\gamma} e^{-k/k_c}$$

where k_c represents a critical connectivity and γ is power-law exponent, both γ and k_c are constants. This seems to be a more widespread observation for social networks not just limited to those of the scientific collaboration type. In fact, several measured social collaboration networks do not follow a power-law degree distribution and are best fitted by the exponentially truncated power-law above (Barabási et al., 2002; Newman, 2001a, 2001d). The value of the cutoff size, k_c, varies considerably. This form is commonly seen in physical systems and suggests an underlying degree distribution that follows a power law, but with some imposed constraint that places a limit on the maximum value of k_c. One possible explanation of this cutoff in the present case is that it arises as a result of the finite time (8-year) window of data used.

The exponent λ of the power-law distribution $P(k)$ is very interesting. The value $\lambda = 2$ forms a dividing line between two fundamentally different behaviors of the network. For networks with $\lambda < 2$, their general properties are dominated by the few actors who have a large number of collaborators, whereas networks with $\lambda > 2$ are dominated by the common actors with few collaborators (Newman, 2003). One can find that in biomedical research, highly connected individuals do not determine the average characteristics of their field, despite their names appearing on a lot of papers. In physics and computer science, on the other hand, it appears that such individuals do determine the characteristics (Newman, 2001a, 2001d). The cumulative data with logarithmic binning in our case indicates $\gamma = 0.25$ and $k_c = 25$. Because the difference of the exponents between scale-free $P(k > n)$ and $P(k)$ is exactly 1, i.e., $\lambda = \gamma + 1$ (Newman, 2003). Here $\lambda = 1.25$, implying that the average properties of the collaboration networks are dominated by the few contractors who have more than 25 collaborators rather than by the many contractors with few collaborators.

3.4. Average path length decreases

In collaboration network, the ability of two nodes (contractors), i and j, to communicate with each other depends on the length of the shortest path, d_{ij}, measured in number of edges between them. The average path length L, also called "average separation of the network", is the average length of the shortest paths between all possible node pairs and, in turn, an indicator of the networks interconnectedness (Wasserman and Faust, 1994). Consider an undirected network:

$$L = \frac{1}{N(N+1)/2} \sum_{i \geq j} d_{ij}$$

where d_{ij} is the shortest path length from node i to node j. N is the total number of nodes in the connected component.

Fig. 8 indicates the evolution of average path length of the giant component of contractors' collaboration network. The average path length L is about 2.0 in 2004 and it grows sharply in the first term, reaching a value of about 4.8 in 2006. It then decreases throughout the second term, reaching a value of 3.97. In a rather speculative manner, values of L before 2007 are not meaningful, as the network was essentially composed of small independent components. Focus on the meaningful values of L after the emergence of giant component, the fact that L decreases might be due to the relative importance of internal edges, i.e., collaboration between present contractors that were previously part of the database. Fig. 9 shows the evolution of the collaboration type in the giant component to a given year. One can find that the edges (collaborations) between two existing contractors (an existing contractor is a contractor who was already present in the network prior to the new edge) are more often than the edges between new contractors and existing

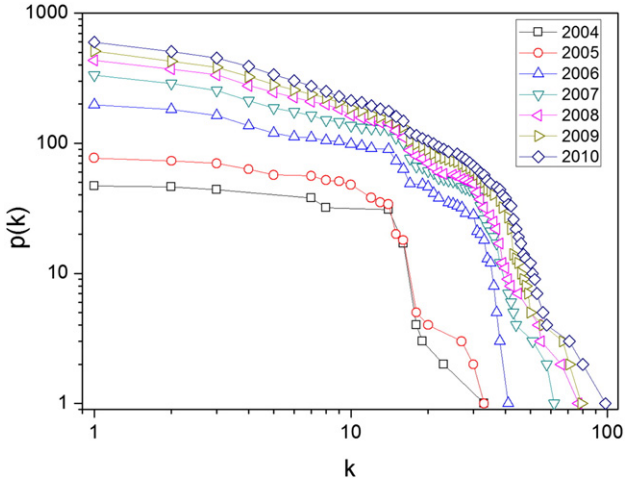


Fig. 7. Evolution of the cumulative degree distribution of the largest component up to the given year.

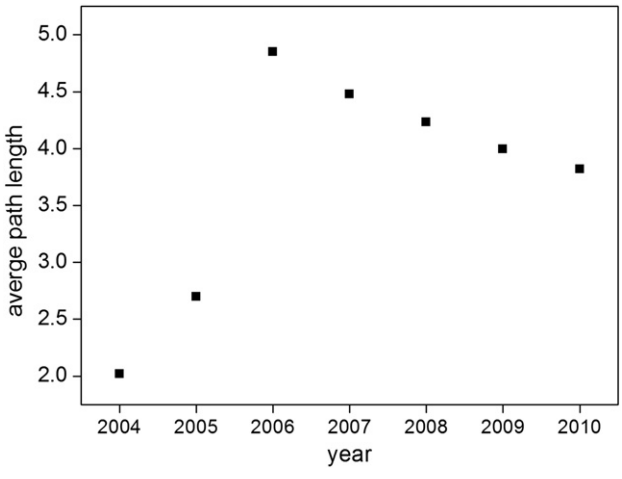


Fig. 8. Evolution of the average path length of the largest component up to the given year.

contractors in the network. The former will tend to increase the interconnectivity thus decreasing the average path length, while the latter should have the opposite effect. It is thus possible that L decreases over time.

The average path length L of the giant component of the contractors' collaboration network until 2010 is 3.97. The appropriate phrase is perhaps "four degrees of separation" for the typical distance between pairs of contractors in NQAPC. It takes an average of only about 4 steps to reach a randomly chosen contractor from any other, which has a profound effect on the way the construction community operates. Low "average path length" between contractors allows for the rapid diffusion of new ideas, technology and resources among members to jointly participate in constructing large-scale construction projects, thus might be regarded as a desirable property of the network.

3.5. Clustering coefficient decays with time

In complex network, clustering (transitivity) measures the average probability that "the neighbor of my neighbor is also my neighbor" (Newman, 2003). This property shows that there exist subgroups (communities) which know each other better than the rest of the network. The definition of clustering can be formalized as follows. Let a connected triple be a triple of nodes x, y, and z such that x is linked to both y and z. That is, y and z have a common neighbor, x. A triangle is a triple of nodes such that all pairs of nodes are connected by an edge. Note that a triangle codes for three connected triples, each one centered at one node of the triangle. The clustering coefficient of a network can be defined as the fraction of connected triples in the graph that are triangles (Newman, 2003)

$$C = \frac{3 \times number\ of\ triangles\ on\ the\ graph}{number\ of\ connected\ triples\ nodes}$$

The range of C is 0 to 1. $C = 1$ for a fully connected network and for a random graph, C tends to zero as $1/N$ as the network becomes large.

Fig. 10 shows the evolution of the clustering coefficient in the cumulative largest component. In the contractors' collaboration network, the clustering coefficient decreases steadily over time, from an initial value of 0.95 (2003), to 0.632 (2010). This suggests that collaboration patterns in the network become more sparse and uniform across the network over time. From 2007, it decreases slightly and tends to reach an asymptotic value of 0.63, although this is a striking observation, it is probably too early to be confident that it will not change in the future. In contrast, Tomassini and Luthi (2007) found that C increases slightly and tends to reach an asymptotic value in their study. A possible explanation of the difference can be found by studying the proportions of the types of new collaborations. According to the definition of clustering, a new contractor working on a project with at least two contractors will join the network with a clustering coefficient of 1, hence necessarily increasing the average clustering coefficient of the network. The majority of the new collaborations are among existing contractors which could indeed explain the increase of the overall clustering coefficient in our case. As can be seen in Fig. 9, among the new collaborations in the cumulative largest component to a given year, more of them are between two existing contractors than between a new contractor and an existing one. Consequently, the network average clustering coefficient in our case will probably tend to decrease.

The clustering coefficient in the contractors' collaboration network tells us how much the collaborators are willing to cooperate with each other. For the contractors' collaboration network until 2010, there is a very strong clustering $C = 0.636$, which means that on average, the chance that two contractors who share a common collaborator cooperated within a project is almost two-thirds. A high clustering coefficient means that construction contractors know each other quite well. A number of explanations of this result are possible. To some extent, it is certainly the result of the appearance of projects involving three or more contractors who cooperated with each other. Such projects clearly contain trios of contractors and led $C = 1$. However, the values measured here cannot be entirely

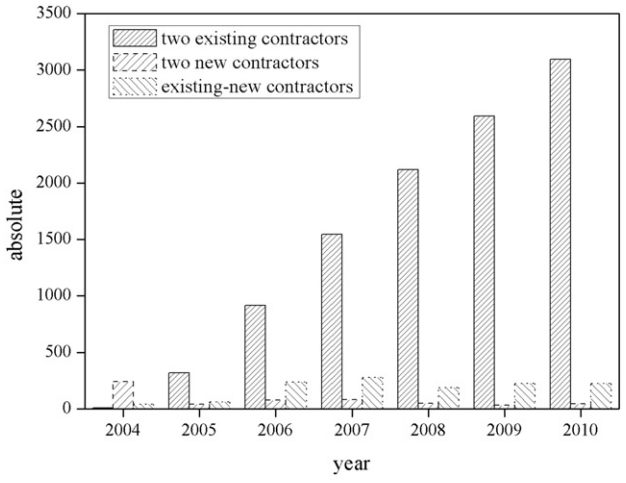


Fig. 9. Evolution of the collaboration type in the largest component up to the given year.

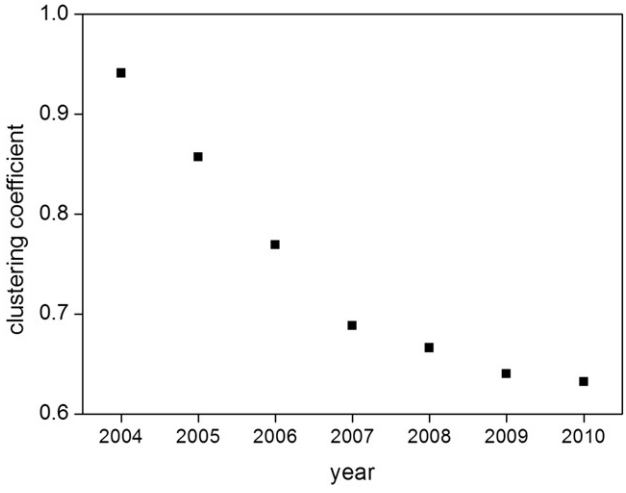


Fig. 10. Evolution of the clustering coefficient of the largest component up to the given year.

accounted for in this way. It may indicate also that contractors either introduce their collaborators to one another, thereby engendering new collaborations, or perhaps that contractors bring sets of collaborators together to form a variety of new collaborations.

4. Discussion

With the aid of social collaboration network approach, the inter-contractors' collaboration networks within the National Quality Award Projects of China (NQAPC) for the period 2003 to 2010 are abstracted. The networks are created with nodes representing Chinese top construction contractors and edges representing collaborations between the contractors. More specifically, the network research mainly focused on the global structure of the network, and the network constructed is large—containing about eight hundred contractors— and for which a reliable statistical description of the structural properties is possible. Some meaningful global network measures are applied to empirically investigate the structural evolution properties of these networks. Table 2 summarizes most statistics for the contractors' collaboration network until 2010, together with those observed for some scientific collaboration networks reported in Newman (2001a).

Most of the interesting features of this study are comparable to those of the larger databases. Some findings and their implications in construction industry are as follows:

1. As the construction industry in China develops, it undergoes a structural transition in the contractors' collaboration structure between a small disconnected network to a much larger network where a giant component of collaboration appears. The giant component size ($s = 0.781$) until 2010 is slightly smaller than analogous values for several scientific collaboration networks which oscillate between 0.8 and 0.92 (e.g., Newman, 2001a, 2001b, 2001c). This indicates that the inter-contractors' collaboration network in China is still growing. There might be independent contractors in smaller component who are likely to become part of the giant component in the next few years. For this to happen, only a single new collaboration between two contractors each belonging to one of the components is needed.
2. The average number of collaborations per contractor in NQAPC is higher than that of the purely theoretical disciplines (3.59 in computer science), less in the wholly or partly experimental ones (18.1 in biomedicine) (Newman, 2001a, 2001b, 2001c). Moreover, despite the diversity of network kind and network scale, a fat-tailed power-law degree distribution is the main property of these networks (Newman, 2003). The degree distributions of the collaboration networks are all well fit by power-law forms with an exponential cutoff. In the contractors' collaboration network, the presence of power-law suggests that there may be "preferential attachment" in which contractors who have previously cooperated are more likely to be parts of new collaborations than those who have not previously cooperated. The power-law distribution exponent $\lambda < 2$ implies that the general properties of the collaboration networks are dominated by the few contractors with many collaborators. Although there are difference of the types and nodes and connections, all the networks in Table 2 have the same power-law distribution degree distribution, which indicates that all the social collaboration networks can be found to have the same structure evolution mechanism.
3. It is easy to find that the collaboration networks exhibit both short average path length and high clustering coefficient. One notable difference is the relative smallness of the average path length in our case. For databases of scientific collaboration that have been studied, such as *NCSTRL* (preprints in computer science) and *Medline* (published papers on biomedical research), which contain more than 10,000 nodes (scientists), the average path lengths are no more than 10 (Newman, 2001a, 2001b, 2001c). The clustering coefficient is rather high, which shows that contractors know each other quite well within the giant component. In contrast in Medline and Mathematics (articles from *Mathematical Reviews*), where scientists from different sub-disciplines seldom collaborate, the clustering coefficient is much lower (e.g., Medline: 0.066; Mathematics: 0.15) (Newman, 2001a, 2001b, 2001c). This may indicate the small-world (Watts and Strogatz, 1998) characteristic of contractors' collaborations within NQAPC. In our cases, the average path length tends to decrease and the clustering coefficient slightly decreases. The former indicates that the collaboration structures become tightly connected and the latter suggests that collaboration structures become more sparse and uniform across the network over time.

Generally speaking, the network analysis in this study mainly focuses on the structure evolution of the inter-contractors collaboration networks, rather than the efficiency and effectiveness of the collaborations. Whether do the interesting

Table 2
Fundamental statistics for some scientific collaboration networks.

Quantity	NQAPC 2010	NCSTRL	SPIRES	Mathematics	Medline
Number of nodes	599	11,994	56,627	253,339	1,520,251
Size of the giant component (%)	78.1	57.2	88.7	82.0	92.6
Average degree	11.215	3.59	173	2.94	18.1
Average path length	3.821	9.7	4.0	7.73	4.6
Clustering coefficient	0.636	0.496	0.726	0.15	0.066

NQAPC 2010 is our network in 2010. NCSTRL is a database of preprints in computer science. SPIRES is a data set of papers in high-energy physics. Mathematics is composed of articles from *Mathematical Reviews*. Medline is a database of articles on biomedical research.

features of this study have positive impact on the information, knowledge and capital circulation in the construction community within NQAPC is a highly debatable question, which can be solved by investigating the correlations between network structure measures and network performance.

5. Conclusion

Understanding the evolution of the collaboration relationships in a construction industry from a network view is a matter of substantial theoretical and practical interest. This study stressed the importance of the global network viewpoint through analyzing real collaboration cases in construction field. The preliminary analysis of the global measures from complex network (average path length, clustering coefficient, and degree distribution) renders us a good understanding of the evolution of the range and configuration of the network over time. Such interesting findings not only would help governors establish the generality of the cognitive and social mechanisms that underlie processes of construction field, but could help contractors to better develop more plausible collaboration strategies, such that they can choose competent partners by identifying network properties and paying attention to the network positioning strategy development. We feel that our results are interesting and useful in the way that it extended previously social network studies in construction field by providing a comprehensive and systematical picture of the principle of how inter-organizational network evolved from an industry perspective.

The most important recognition is that the structural evolution of the contractors' collaboration networks possesses some kind of 'universality' for which other collaboration networks exhibit. For example, it is interesting that the power-law degree distribution with an exponential cutoff holds, which suggests that contractors' collaboration progress in China can be shown to a certain level of self-organization based on rules of preferential attachment with social constraints. This dynamic organization presents particular challenges to policymaking in the construction community. As self-organized system, the overall structure of contractors' collaboration network is formed through the individual interests of contractors seeking profits and reputation. Deterministic and rigid attempts at directing it are destined to fail. From a construction market management viewpoint (government), this result is important. It can provide indications on how to optimize some performance, for example, the most effective attitude is an adaptive one, requiring a flexible policy for changing it swiftly to the transformation and development of the whole construction community.

Moreover, in our study, the collaboration network structure (e.g., scale-free) is comprehensively the reflection of the collaboration behavior, such as the efforts and motivations of bridging knowledge and expertise gaps, sharing of the burdens of risk exposure, and exploring local market opportunities (Walker and Johannes, 2003) of individual contractors. These collaboration behaviors are based on regional preferential mechanism (contractors at the same area are more likely to collaborate), professional preferential mechanism (contractors belong to the same professional field are more likely to collaborate) and social capital preferential mechanism (contractors who have previously cooperated are more likely to collaborate). And despite the diversity of network kind and network scale, this preferential mechanism may be found in the collaboration networks from other countries' construction field and lead to homogeneity of network structure, i.e., scale-free characteristics (e.g., Newman, 2003). Isomorphism is a "constraining process that forces one unit in a population to resemble other units that face the same set of environmental conditions" (DiMaggio and Powell, 1983). We called the network evolution process under certain path selection and evolutionary regularity as network isomorphism. This is a very interesting remark, and needs further consolidation.

Despite the advantages, the results of this study must be viewed with respect to a number of limitations. First, the findings do not perfectly fit to every country owing to the limited data samples in China. Analysis should be performed by reflecting specific countries' construction market. Second, the study is limited to collaborations among contractors in domestic construction ventures. Existing literatures have suggested that international collaboration between local contractors and oversea contractors tends to increase when overseas construction projects are concerned (e.g., Park et al., 2011). Future research could be outspread in analyzing collaborations between domestic contractors and foreign contractors across different cultures. Third, the contractors in construction field are suspended in multiple and diverse webs of relationships, and we are likely to investigate the overall network patterns of different relationships. For example, as the interpersonal Guanxi network (Farh et al., 1998) can be seen as a complement to the formal network, the similarities and differences between the Guanxi based network and the contractual based network, need to be explored and exploited in depth in future research, which may provide a deeper understanding of the collaboration behaviors among contractors in construction industry.

Conflict of interest

The authors have declared that no conflict of interest exists.

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